

Legislative Coordination and Institutional Change

A Bipartite Network Analysis of Coalition Dynamics in the Brazilian Chamber of Deputies

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| The puzzle

- ◆ Polarization is supposed to make legislatures **ungovernable**.
- ◆ Brazil, 2014–2016: impeachment, Lava Jato, deep recession, record polarization.
- ◆ And yet the Chamber **kept assembling majorities**, across an impeachment and a change of power.
- ◆ **How? And at what cost?**

Coalitional presidentialism, and the 2014–16 inversion

- ◆ Real presidential powers, no majority party: to govern is to **assemble a multiparty coalition** (Abranches 1988).
- ◆ Held by the president's tools (budget, posts, agenda); the **Centrão** sells support as *agent*.
- ◆ 2014–16 weakens the presidency while budget reform moves its main tool to Congress: *impositive* amendments (2015).
- ◆ The relation **inverts**: the pivot stops needing favor and begins to *set* terms (*governo congressual*).

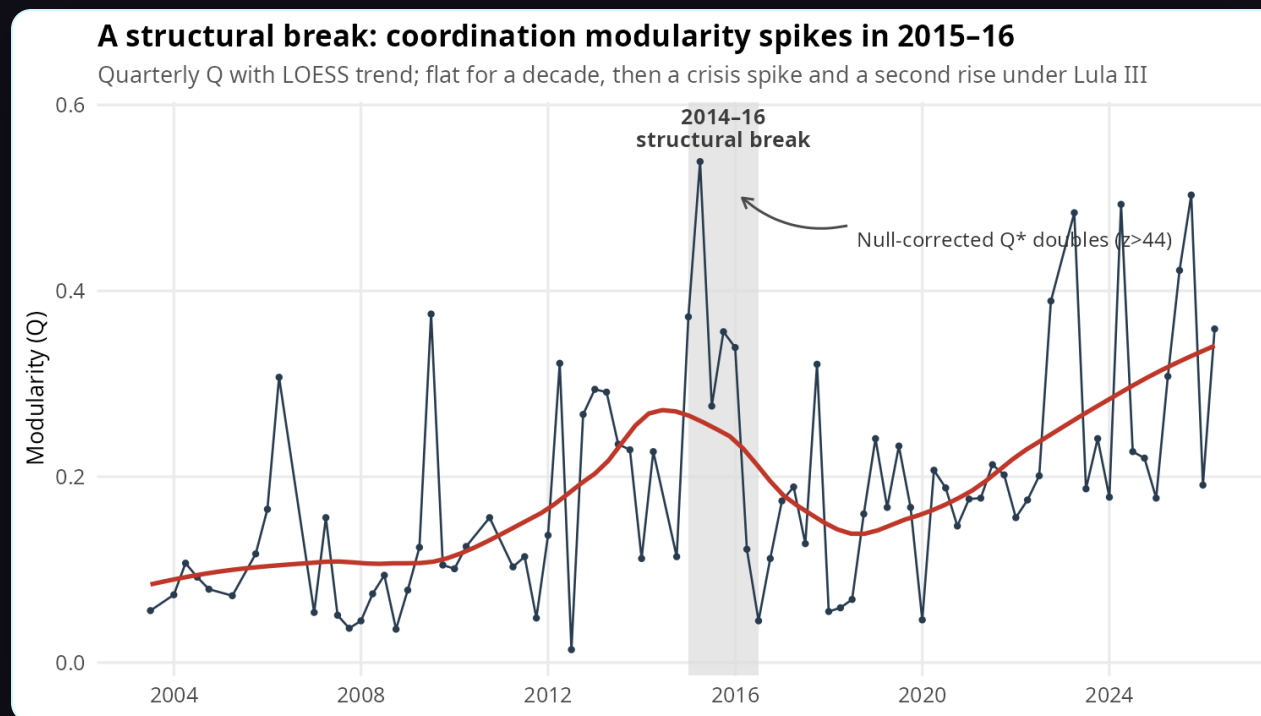
| The thesis, in one line

- ◆ Polarization did not eliminate Brazil's pragmatic center; it made that center the pivot through which every government assembles majorities.
- ◆ Coordination runs by **position relative to the executive of the day**, not by a fixed ideological axis.
- ◆ The Centrão is **old**; what changed is its **position**: disciplined *agent* → autonomous *pivot*.
- ◆ Backdrop: the 2015 shift to mandatory ("impositive") budget amendments, part of a broader transfer of budgetary power to Congress.

| Why networks, not ideal points

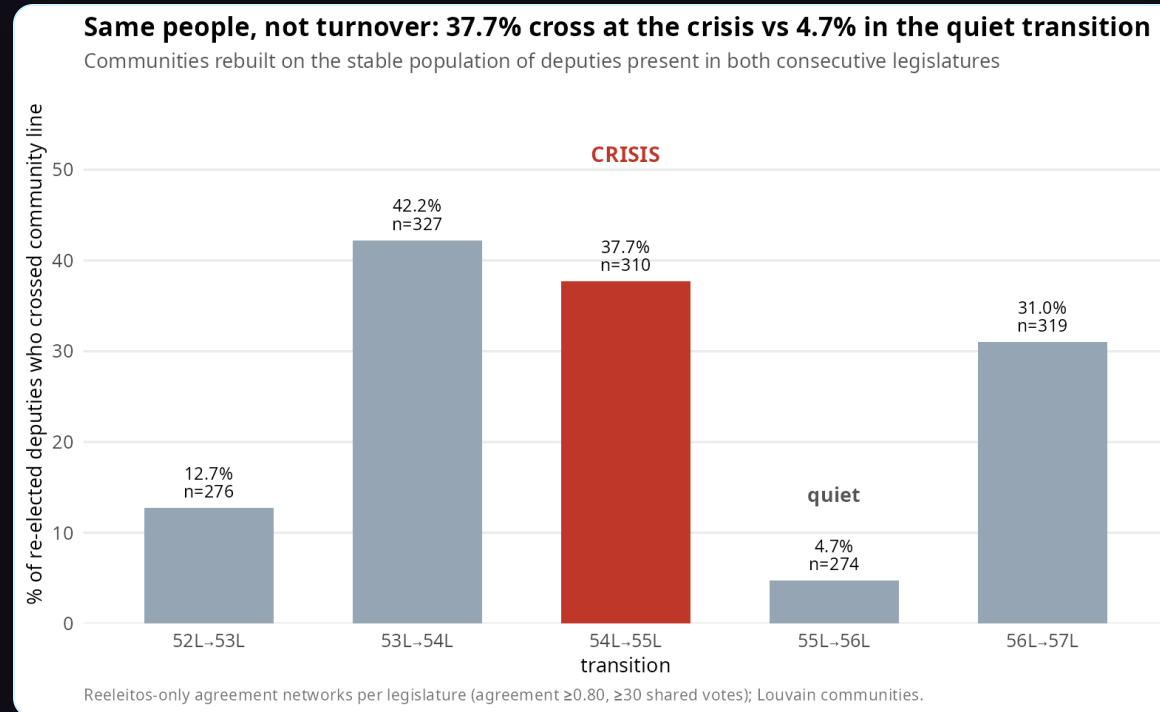
- ◆ Bipartite incidence: **deputies × bills** → **deputy–deputy agreement graph** → **community detection**.
- ◆ Ideal points assume ideology organizes voting and compress change onto one left–right axis.
- ◆ A bloc that **recomposes** while ideology looks stable is what an axis **cannot see**.
- ◆ The network makes no axial assumption: the shift shows up as **bloc recomposition**.

A structural break: modularity spikes in 2015–16



Quarterly modularity (Q): flat ~0.15, spiking to 0.51 in 2016Q1. Null-corrected Q^* doubles at the 55th and 57th legislatures.

Same people, not turnover: 37.7% cross vs 4.7%

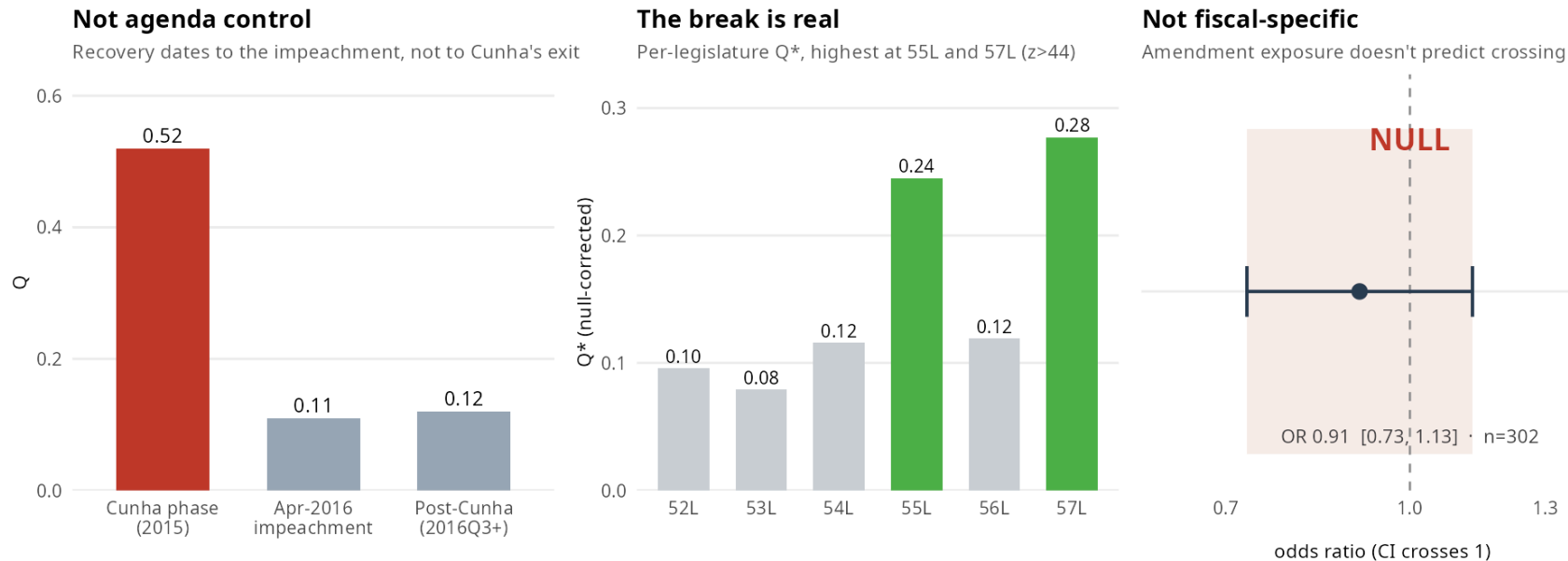


Reelected deputies crossing community lines: 37.7% at the 2014–15 break, against 4.7% in the quiet transition that follows.

Not turnover. Not agenda. Not just fiscal.

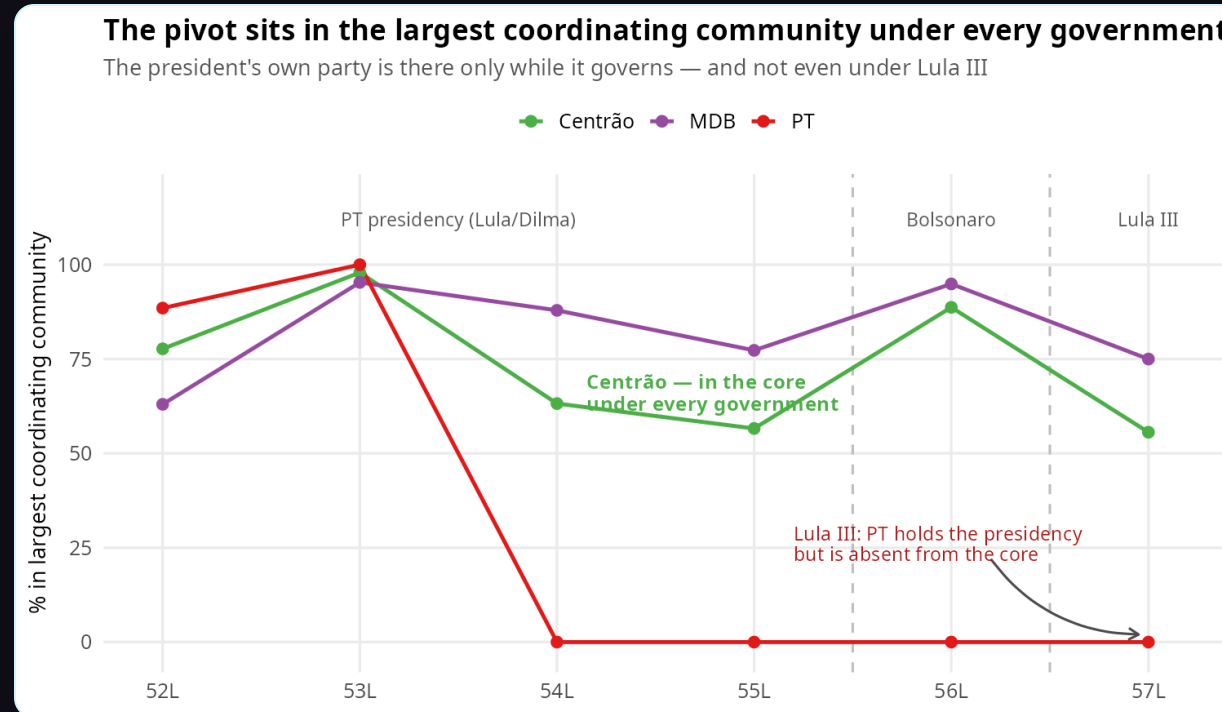
The reorganization survives the observable rivals

Collinear shocks preclude estimating one effect; the eliminative design kills each rival in turn.



A: the Q-recovery dates to the impeachment, not Cunha's removal. B: null-corrected Q*, sharpest at the 55L and 57L. C: the null fiscal dose-response. The break survives every rival; the shocks are collinear, so the design eliminates rivals rather than estimating one effect.

The pivot: in the largest community under every government



Share of each party in the largest coordinating community. The Centrão is present under every government; the PT only while it governs, not even under Lula III.

| From agent to principal, and the cost

- ◆ Budgetary transfers (mandatory "impositive" amendments from 2015; the rapporteur-controlled "secret budget" a **later consolidation** from 2020) **reverse the relation**: the pivot begins to **set the terms** ("governo congressual").
- ◆ Governability comes to be **sold, not represented**.
- ◆ The cost is **accountability**, not gridlock: a bloc available to any government is hard to hold to account.

| Contribution, and what the case teaches

- ◆ **Measurement:** a longitudinal, network-based quantity for a *governista* pivot's position, persistence, and decision weight.
- ◆ **Design:** an eliminative chain for **collinear shocks**, where temporal network models are unavailable.
- ◆ **Not only Brazil:** polarization *raised* the value of a low-programmatic office-seeking pivot (cousins in Israel, post-war Italy), a mode of party-system change without collapse or loss of discipline.

The pivot, not the program, is what keeps it governable.

Thank you

Questions welcome. Backup slides follow.

Andéliton Soares

Method

- ◆ Pairwise Krehbiel-style agreement on co-voted bills; edge filter on agreement and shared-vote count.
- ◆ Community detection: **Louvain** (Leiden interchangeable, $ARI \approx 0.98-1.00$).
- ◆ Q^* : observed modularity minus mean of a within-votation permutation null (150–200 draws); $z > 44$, $p < 0.001$.
- ◆ Within-deputy panel: reelected members across consecutive legislatures, matched by majority overlap.

Robustness

- ◆ **Thresholds:** 12-cell grid (edge 0.70–0.85 × shared-vote 20–40); 54L → 55L crossing in [29.6%, 44.8%].
- ◆ **Projection:** Jaccard and resource-allocation robust; collapses only under Newman (no community structure).
- ◆ **Algorithms:** Louvain, Leiden, Infomap, Walktrap; crisis transition least similar on label-free ARI/NMI.
- ◆ **Near-unanimous votes** dropped (modal > 95/90/85%): crossing holds at 30.6 / 28.6 / 26.0%.

| Why not a temporal ERGM

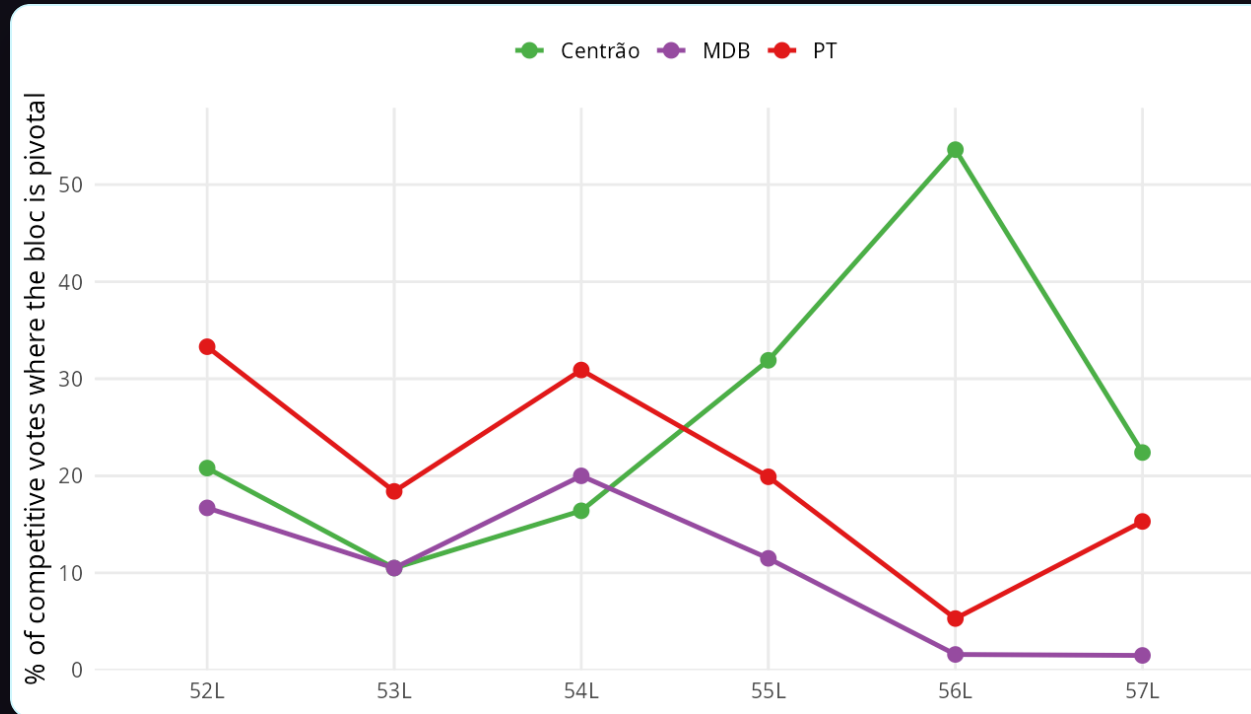
- ◆ Given collinear shocks and poor mixing in the bipartite specification, a bootstrap-TERGM with the amendment reform as covariate did **not provide a reliable route to the estimand** here.
- ◆ OLS-DiD: the reform's coefficient on modularity **collapses** under controls; variance absorbed by the Cunha dummy.
- ◆ Static bipartite ERGM on the 54L did **not mix** after 137 minutes (estimation degenerated).
- ◆ Hence the eliminative design: an **informative outcome**, not a hidden failure.

| The Centrão, defined

BACKUP

- ◆ A reproducible, sourced, **dated** set of mid-sized, low-programmatic, *governista* parties (Testa et al. 2024).
- ◆ Each party enters only from the legislature it **joined the governista orbit** (e.g., DEM from the 55th).
- ◆ PSB coded left (out); MDB tracked separately as a pivot.
- ◆ Survives **six** membership schemes; the bloc grew from **19.5%** to **66.1%** of the chamber.

Decision weight migrates from PT to Centrão



On competitive roll-calls, the decisive core migrates from the PT (while it governs) to the Centrão (net of the president's party).